

12.2 Confidence ellipsoids

Two 95% intervals, one for each of two coefficients, do not make a 95% statement about the pair, and they ignore the correlation between the estimates. The natural joint region is an ellipsoid: the set of values the F test does not reject, shaped like the covariance matrix of the estimates, with shadows that are wider than the t intervals. This geometry explains an F test that rejects while every t test accepts, and the reverse.

12.2.1 The ellipsoid

Let $\mathbf{\Lambda}$ be a $p \times q$ matrix of rank q whose columns lie in $\mathcal{C}(\mathbf{X}^\top)$, so that every entry of $\boldsymbol{\phi} = \mathbf{\Lambda}^\top \boldsymbol{\beta}$ is estimable. Put

$$\hat{\boldsymbol{\phi}} = \mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}}, \quad \mathbf{W} = \mathbf{\Lambda}^\top \mathbf{G} \mathbf{\Lambda}.$$

By Theorem 9.2.4(a), neither depends on the choices of $\hat{\boldsymbol{\beta}}$ and $\mathbf{G}$, and $\mathbf{W}$ is positive definite. By Theorem 7.5.1(a), $\hat{\boldsymbol{\phi}} \sim N_q(\boldsymbol{\phi}, \sigma^2 \mathbf{W})$.

Theorem 12.2.1 — Confidence ellipsoid for $\mathbf{\Lambda}^\top \boldsymbol{\beta}$

With the notation above, let $F_\alpha = F_\alpha(q, n - r)$ and

$$E(\mathbf{Y}) = \{\boldsymbol{\phi}_0 \in \mathbb{R}^q : (\hat{\boldsymbol{\phi}} - \boldsymbol{\phi}_0)^\top \mathbf{W}^{-1}(\hat{\boldsymbol{\phi}} - \boldsymbol{\phi}_0) \leq q s^2 F_\alpha\}. \quad (12.2.1)$$

- (a) $E(\mathbf{Y})$ is an exact $1 - \alpha$ confidence set for $\mathbf{\Lambda}^\top \boldsymbol{\beta}$.
- (b) $E(\mathbf{Y})$ is a solid ellipsoid centred at $\hat{\boldsymbol{\phi}}$. If $\mathbf{W} = \sum_{k=1}^q w_k \mathbf{u}_k \mathbf{u}_k^\top$ is a spectral decomposition, its principal axes point along $\mathbf{u}_1, \dots, \mathbf{u}_q$ and have half-lengths $\sqrt{q s^2 F_\alpha w_k}$. Its volume is

$$\frac{\pi^{q/2}}{\Gamma(q/2 + 1)} (q s^2 F_\alpha)^{q/2} \det(\mathbf{W})^{1/2}.$$

- (c) (*Duality with the F test.*) For $\mathbf{d} \in \mathbb{R}^q$, let $\text{SSE}_{\mathbf{d}}$ be the minimum of $\|\mathbf{Y} - \mathbf{X}\mathbf{b}\|^2$ over $\{\mathbf{b} : \mathbf{\Lambda}^\top \mathbf{b} = \mathbf{d}\}$ and let

$$F_{\mathbf{d}} = \frac{(\text{SSE}_{\mathbf{d}} - \text{SSE})/q}{s^2} = \frac{(\hat{\boldsymbol{\phi}} - \mathbf{d})^\top \mathbf{W}^{-1}(\hat{\boldsymbol{\phi}} - \mathbf{d})}{q s^2}.$$

Then $\mathbf{d} \in E(\mathbf{Y})$ iff $F_{\mathbf{d}} \leq F_\alpha$, that is, iff the level- α F test of $H : \mathbf{\Lambda}^\top \boldsymbol{\beta} = \mathbf{d}$ does not reject.

Proof.

- (a) Since $\mathbf{W}$ is invertible, $\mathbf{W}^{-1}$ is its only generalized inverse, and Corollary 7.5.3 with $\text{rank}(\mathbf{\Lambda}) = q$ gives

$$\frac{(\hat{\boldsymbol{\phi}} - \boldsymbol{\phi})^\top \mathbf{W}^{-1}(\hat{\boldsymbol{\phi}} - \boldsymbol{\phi})}{q s^2} \sim F(q, n - r)$$

at every parameter value. This is a pivot, and $\boldsymbol{\phi} \in E(\mathbf{Y})$ is the event that it is at

most F_α .

- (b) Put $c = q s^2 F_\alpha$ and write $\phi_0 - \hat{\phi} = \sum_k c_k \mathbf{u}_k$. Since $\mathbf{W}^{-1} = \sum_k w_k^{-1} \mathbf{u}_k \mathbf{u}_k^\top$, the defining inequality becomes $\sum_k c_k^2 / w_k \leq c$, the standard form of an ellipsoid with half-axes $\sqrt{c w_k}$ along $\mathbf{u}_k$. Equivalently $E(\mathbf{Y}) = \hat{\phi} + \mathbf{W}^{1/2} B$, where B is the ball of radius $\sqrt{c}$ in $\mathbb{R}^q$. A linear map multiplies volume by the absolute value of its determinant, so the volume is $\det(\mathbf{W})^{1/2} c^{q/2}$ times the volume $\pi^{q/2} / \Gamma(q/2 + 1)$ of the unit ball.
- (c) The two expressions for F_d agree by Theorem 9.2.4(c), and the equivalence is the definition of $E(\mathbf{Y})$.

■

The statistic F_d is the F statistic of the general linear hypothesis in both of its forms (see Theorem 11.3.2), and part (c) is Proposition 12.1.2 in concrete form. In particular the test of $\mathbf{\Lambda}^\top \boldsymbol{\beta} = \mathbf{0}$ rejects iff the ellipsoid misses the origin.

In the full-rank case with $\mathbf{\Lambda} = \mathbf{I}_p$, the region for the whole coefficient vector is

$$\begin{aligned} \{\mathbf{b} : (\hat{\boldsymbol{\beta}} - \mathbf{b})^\top \mathbf{X}^\top \mathbf{X} (\hat{\boldsymbol{\beta}} - \mathbf{b}) &\leq p s^2 F_\alpha(p, n - p)\} \\ &= \{\mathbf{b} : \|\mathbf{X} \hat{\boldsymbol{\beta}} - \mathbf{X} \mathbf{b}\|^2 \leq p s^2 F_\alpha(p, n - p)\}. \end{aligned}$$

The second form says that $\mathbf{b}$ is in the region iff its fitted vector $\mathbf{X} \mathbf{b}$ is close to the least squares fit $\hat{\mathbf{Y}}$. This suggests looking at the mean vector directly.

12.2.2 A ball in the column space

Proposition 12.2.2 — Confidence ball for the mean vector

Let $\boldsymbol{\mu} = \mathbf{X} \boldsymbol{\beta}$ and $c_r = r s^2 F_\alpha(r, n - r)$. The set

$$B(\mathbf{Y}) = \{\boldsymbol{\mu}_0 \in \mathcal{C}(\mathbf{X}) : \|\hat{\mathbf{Y}} - \boldsymbol{\mu}_0\|^2 \leq c_r\}$$

is an exact $1 - \alpha$ confidence set for $\boldsymbol{\mu}$: a ball of radius $\sqrt{c_r}$ in the r -dimensional space $\mathcal{C}(\mathbf{X})$, centred at the fitted vector. Moreover, if $\mathbf{\Lambda} = \mathbf{X}^\top \mathbf{T}$ for an $n \times q$ matrix $\mathbf{T}$ (such a $\mathbf{T}$ exists by Theorem 8.3.1), the image of $B(\mathbf{Y})$ under $\boldsymbol{\mu}_0 \mapsto \mathbf{T}^\top \boldsymbol{\mu}_0$ is the set

$$\{\phi_0 : (\hat{\phi} - \phi_0)^\top \mathbf{W}^{-1} (\hat{\phi} - \phi_0) \leq c_r\},$$

which contains the ellipsoid [Eq. 12.2.1](#).

Proof. $\hat{\mathbf{Y}} - \boldsymbol{\mu} = \mathbf{M}(\mathbf{Y} - \mathbf{X} \boldsymbol{\beta})$, and $\mathbf{M}$ is symmetric idempotent of rank r , so $\|\hat{\mathbf{Y}} - \boldsymbol{\mu}\|^2 / \sigma^2 \sim \chi^2(r)$ by Theorem 4.3.2(a), independently of SSE (Theorem 7.5.1(c)). Dividing by $r s^2 / \sigma^2$ gives an $F(r, n - r)$ pivot, and the first claim follows as before.

For the image, put $\mathbf{A} = \mathbf{T}^\top \mathbf{M}$, a $q \times n$ matrix. For $\boldsymbol{\mu}_0 \in \mathcal{C}(\mathbf{X})$, $\mathbf{T}^\top \boldsymbol{\mu}_0 = \mathbf{A} \boldsymbol{\mu}_0$, and $\mathbf{A} \mathbf{A}^\top = \mathbf{T}^\top \mathbf{M} \mathbf{T} = \mathbf{W}$ (Theorem 9.2.4(a)). Also $\hat{\phi} = \mathbf{T}^\top \hat{\mathbf{Y}} = \mathbf{A} \hat{\mathbf{Y}}$. So the image consists of the points $\hat{\phi} - \mathbf{A} \mathbf{u}$ with $\mathbf{u} \in \mathcal{C}(\mathbf{X})$ and $\|\mathbf{u}\|^2 \leq c_r$. If $\mathbf{z} = \mathbf{A} \mathbf{u}$ with $\|\mathbf{u}\|^2 \leq c_r$, then $\mathbf{z}^\top \mathbf{W}^{-1} \mathbf{z} = \mathbf{u}^\top \mathbf{A}^\top \mathbf{W}^{-1} \mathbf{A} \mathbf{u} \leq \|\mathbf{u}\|^2 \leq c_r$, because $\mathbf{A}^\top \mathbf{W}^{-1} \mathbf{A}$ is symmetric and idempotent, hence a projection, and a projection does not increase length (Proposition 6.3.4(d)). Conversely, if $\mathbf{z}^\top \mathbf{W}^{-1} \mathbf{z} \leq c_r$, then $\mathbf{u} = \mathbf{A}^\top \mathbf{W}^{-1} \mathbf{z}$ lies in

$\mathcal{C}(\mathbf{M}) = \mathcal{C}(\mathbf{X})$, satisfies $\mathbf{A}\mathbf{u} = \mathbf{z}$, and has $\|\mathbf{u}\|^2 = \mathbf{z}^\top \mathbf{W}^{-1} \mathbf{z} \leq c_r$.

For the containment it remains to show $qF_\alpha(q, \nu) \leq rF_\alpha(r, \nu)$ when $q \leq r$. Let $U_q \sim \chi^2(q)$, $U' \sim \chi^2(r-q)$ and $V \sim \chi^2(\nu)$ be independent. Then $U_q + U' \sim \chi^2(r)$, so for every c , $\Pr\{(U_q + U')/(V/\nu) \leq c\} \leq \Pr\{U_q/(V/\nu) \leq c\}$. The upper α point of the variable on the left is $rF_\alpha(r, \nu)$, and that of the one on the right is $qF_\alpha(q, \nu)$. ■

So all confidence ellipsoids are views of one ball around $\hat{\mathbf{Y}}$ in the column space. Each linear summary $\mathbf{\Lambda}^\top \boldsymbol{\beta}$ maps the ball to an ellipsoid, and since the ball covers $\boldsymbol{\mu}$ with probability $1 - \alpha$, all the images cover their targets *simultaneously* with probability at least $1 - \alpha$, at the price of the constant $rF_\alpha(r, \nu)$ in place of $qF_\alpha(q, \nu)$. Section 12.4 takes $q = 1$ and all directions at once.

12.2.3 Shadows

One inequality gives the extent of an ellipsoid in every direction.

Lemma 12.2.3 — Extent of an ellipsoid

Let $\mathbf{W}$ be a $q \times q$ positive definite matrix and $c > 0$. For every $\mathbf{a} \in \mathbb{R}^q$,

$$\max\{\mathbf{a}^\top \mathbf{u} : \mathbf{u}^\top \mathbf{W}^{-1} \mathbf{u} \leq c\} = \sqrt{c \mathbf{a}^\top \mathbf{W} \mathbf{a}},$$

attained at $\mathbf{u} = \sqrt{c/(\mathbf{a}^\top \mathbf{W} \mathbf{a})} \mathbf{W} \mathbf{a}$ when $\mathbf{a} \neq \mathbf{0}$. Consequently $\mathbf{u}^\top \mathbf{W}^{-1} \mathbf{u} \leq c$ iff $|\mathbf{a}^\top \mathbf{u}| \leq \sqrt{c \mathbf{a}^\top \mathbf{W} \mathbf{a}}$ for every $\mathbf{a} \in \mathbb{R}^q$.

Proof. The bound is Corollary 1.7.8(c) with $\mathbf{B} = \mathbf{W}^{-1}$; directly, write $\mathbf{a}^\top \mathbf{u} = (\mathbf{W}^{1/2} \mathbf{a})^\top (\mathbf{W}^{-1/2} \mathbf{u})$. By the Cauchy–Schwarz inequality, $(\mathbf{a}^\top \mathbf{u})^2 \leq (\mathbf{a}^\top \mathbf{W} \mathbf{a})(\mathbf{u}^\top \mathbf{W}^{-1} \mathbf{u}) \leq c \mathbf{a}^\top \mathbf{W} \mathbf{a}$, with equality when $\mathbf{W}^{-1/2} \mathbf{u}$ is a nonnegative multiple of $\mathbf{W}^{1/2} \mathbf{a}$ and $\mathbf{u}^\top \mathbf{W}^{-1} \mathbf{u} = c$, which is the stated $\mathbf{u}$. For the last statement, “only if” follows from the maximum applied to $\mathbf{a}$ and $-\mathbf{a}$. For “if”, take $\mathbf{a} = \mathbf{W}^{-1} \mathbf{u}$: then $(\mathbf{u}^\top \mathbf{W}^{-1} \mathbf{u})^2 \leq c \mathbf{u}^\top \mathbf{W}^{-1} \mathbf{u}$, so $\mathbf{u}^\top \mathbf{W}^{-1} \mathbf{u} \leq c$. ■

Corollary 12.2.4 — Shadows of the confidence ellipsoid

For every $\mathbf{a} \in \mathbb{R}^q$, the projection of $E(\mathbf{Y})$ onto the line spanned by $\mathbf{a}$ is the interval

$$\mathbf{a}^\top \hat{\boldsymbol{\phi}} \pm \sqrt{q F_\alpha(q, n-r)} \text{se}(\mathbf{a}^\top \hat{\boldsymbol{\phi}}), \quad \text{se}(\mathbf{a}^\top \hat{\boldsymbol{\phi}}) = s \sqrt{\mathbf{a}^\top \mathbf{W} \mathbf{a}},$$

and $E(\mathbf{Y})$ is the intersection of the slabs $\{\boldsymbol{\phi}_0 : |\mathbf{a}^\top \hat{\boldsymbol{\phi}} - \mathbf{a}^\top \boldsymbol{\phi}_0| \leq \sqrt{q F_\alpha} \text{se}(\mathbf{a}^\top \hat{\boldsymbol{\phi}})\}$ over all $\mathbf{a}$. Hence, with probability exactly $1 - \alpha$, every one of these intervals contains $\mathbf{a}^\top \mathbf{\Lambda}^\top \boldsymbol{\beta}$.

Proof. Apply Lemma 12.2.3 to $\mathbf{u} = \boldsymbol{\phi}_0 - \hat{\boldsymbol{\phi}}$ with $c = q s^2 F_\alpha$. The extreme values of $\mathbf{a}^\top \boldsymbol{\phi}_0$ over $E(\mathbf{Y})$ are $\mathbf{a}^\top \hat{\boldsymbol{\phi}} \pm \sqrt{c \mathbf{a}^\top \mathbf{W} \mathbf{a}}$, and the set of values in between is attained because $E(\mathbf{Y})$ is convex. The intersection statement is the last part of the lemma. The final claim holds because the event “all intervals cover” is the event $\boldsymbol{\phi} \in E(\mathbf{Y})$.

Example. (A joint region for two coefficients)

In the murder-rate regression of Example (Murder rates: coefficients and a mean), take $\mathbf{A}$ to pick out the coefficients of poverty and single parenthood, so $q = 2$, $\hat{\phi} = (0.2538, 0.3980)^\top$, and the correlation of the two estimates is -0.639 . The negative correlation reflects the positive correlation of the regressors: the data can credit murder to either, but not to both at once. With $F_{0.05}(2, 46) = 3.200$, the 95% ellipse is drawn in Figure 12.2.1. The F statistic for $\phi = \mathbf{0}$ is 39.81, so the origin is far outside.

The shadows of the ellipse on the two axes are $(0.018, 0.490)$ and $(0.188, 0.608)$, wider than the t intervals $(0.066, 0.442)$ and $(0.231, 0.565)$ by the factor $\sqrt{2 \times 3.200}/2.013 = 1.257$. Two points show how the ellipse and the box disagree.

- $\mathbf{A} = (0.442, 0.565)$, the upper corner of the box. Neither t test rejects, but $\mathbf{A}$ raises both coefficients at once, against the correlation, and the joint test rejects: $F = 11.230$, $p = 0.0001$.
- $\mathbf{B} = (0.463, 0.224)$, on the long axis of the ellipse. Both t statistics exceed 2.013 (2.242 and 2.098), but $\mathbf{B}$ trades one coefficient for the other, which the data cannot resolve, and $F = 2.888 < 3.200$.

The ellipse and the box have similar areas (0.1197 and 0.1255) but different shapes. In 40000 simulated data sets from the fitted model the ellipse covered the true pair in a proportion 0.9500 and the box in only 0.9164.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats

data = sm.datasets.statecrime.load_pandas().data
data = data.drop(index="District of Columbia")
y = data["murder"].to_numpy()
X = np.column_stack([np.ones(len(y)), data[["poverty", "single", "urban"]]])
n, p = X.shape
C = np.linalg.inv(X.T @ X)
beta_hat = C @ X.T @ y
s2 = np.sum((y - X @ beta_hat) ** 2) / (n - p)

L = np.eye(p)[1:, [1, 2]]
coefficients
q = L.shape[1]
phi_hat = L.T @ beta_hat
W = L.T @ C @ L
Cov(phi_hat) / sigma^2
```

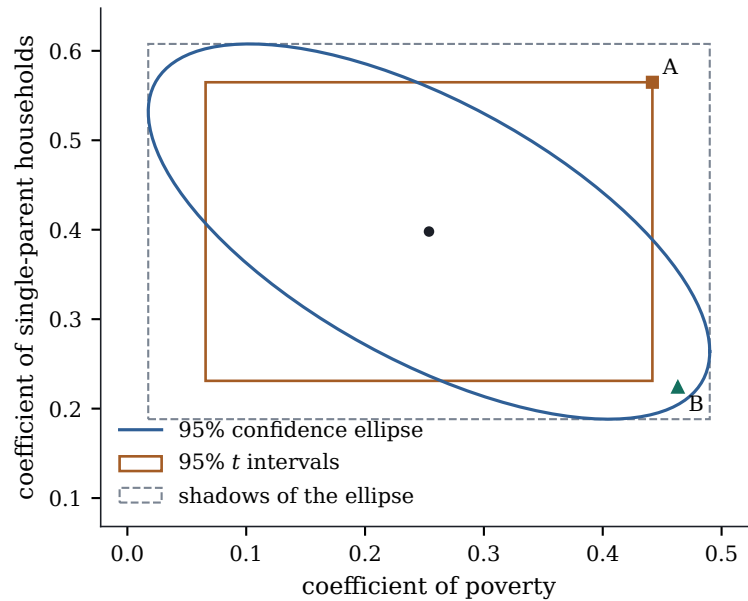


Figure 12.2.1: The 95% confidence ellipse for the poverty and single-parent coefficients, the box of 95% t intervals, and the ellipse's shadows (dashed). A is inside the box but outside the ellipse; B the reverse.

```
Fq = stats.f.ppf(0.95, q, n - p)

def F_stat(d):
    """F statistic for the hypothesis  $\Lambda^T \beta = d$ ."""
    u = phi_hat - d
    return u @ np.linalg.solve(W, u) / (q * s2)

def in_ellipse(d):
    return F_stat(d) <= Fq

corr = W[0, 1] / np.sqrt(W[0, 0] * W[1, 1])
print(f"estimates {phi_hat}, correlation of the estimates {corr:.3f}")
print(f"F({q}, {n - p}) 95% point {Fq:.3f}; F statistic for (0, 0): {F_stat(np.zeros(2)):.2f}")
```

[Run this code online](#)

```
tq = stats.t.ppf(0.975, n - p)
se = np.sqrt(s2 * np.diag(W))
t_box = np.column_stack([phi_hat - tq * se, phi_hat + tq * se])          # one-at-
    a-time t intervals
k = np.sqrt(q * Fq)
shadow = np.column_stack([phi_hat - k * se, phi_hat + k * se])        #
    projections of the ellipse
print("t intervals:\n", t_box.round(3))
print("shadows of the ellipse:\n", shadow.round(3), f"\nwider by the factor {k / tq:.3f}")
```

[Run this code online](#)

12.2.4 Two cautions

The ellipsoid is for a question fixed in advance. If the coefficients are chosen because they look interesting, the ball of [Proposition 12.2.2](#) or the methods of [Chapter 13](#) are the honest tools.

Intervals after a test. If intervals are reported only when a preliminary test of $\mathbf{\Lambda}^\top \boldsymbol{\beta} = \mathbf{0}$ rejects, their coverage *given* the rejection is no longer $1 - \alpha$, and can be larger or smaller (Olshen 1973).

12.2.5 Exercises

12.2.5.1 A. Check your understanding

Exercise A1.

Show that for $q = 1$ the ellipsoid [Eq. 12.2.1](#) is the t interval [Eq. 12.1.1](#). Which fact about the $F(1, \nu)$ distribution does this use?

Solution. With $q = 1$, $\mathbf{W} = w$ is a positive number and the region is $(\hat{\phi} - \phi_0)^2 \leq s^2 w F_\alpha(1, n - r)$. If $T \sim t(\nu)$, then $T^2 \sim F(1, \nu)$ (Definition 4.1.3 and the remark after it), so $\Pr\{T^2 \leq t_{\nu, \alpha/2}^2\} = \Pr\{|T| \leq t_{\nu, \alpha/2}\} = 1 - \alpha$ and $F_\alpha(1, \nu) = t_{\nu, \alpha/2}^2$. The region is $|\hat{\phi} - \phi_0| \leq t_{\nu, \alpha/2} s \sqrt{w}$.

Exercise A2.

For the straight line $E Y_i = \beta_0 + \beta_1 x_i$, write the 95% confidence ellipse for (β_0, β_1) in terms of n , $\bar{x}$ and $\sum x_i^2$. Show that in the centred parameterization $E Y_i = \gamma_0 + \beta_1(x_i - \bar{x})$ the axes of the ellipse for (γ_0, β_1) are parallel to the coordinate axes. When is it a circle?

Solution. $\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} n & n\bar{x} \\ n\bar{x} & \sum x_i^2 \end{pmatrix}$, and the region is $n(\hat{\beta}_0 - b_0)^2 + 2n\bar{x}(\hat{\beta}_0 - b_0)(\hat{\beta}_1 - b_1) + \sum x_i^2(\hat{\beta}_1 - b_1)^2 \leq 2s^2 F_\alpha(2, n - 2)$. In the centred parameterization the cross-product matrix is $\text{diag}(n, S_{xx})$, so the quadratic form has no cross term, and the ellipse has axes along the coordinates with half-lengths $\sqrt{2s^2 F_\alpha/n}$ and $\sqrt{2s^2 F_\alpha/S_{xx}}$. It is a circle iff $S_{xx} = n$, that is, iff the x_i have mean square deviation one.

12.2.5.2 B. Practice

Exercise B1.

Take $q = 2$ and let ρ be the correlation of $\hat{\phi}_1$ and $\hat{\phi}_2$. Show that the area of the ellipse divided by the area of the box of the two t intervals is

$$\frac{\pi F_\alpha(2, n - r)}{2 t_{n-r, \alpha/2}^2} \sqrt{1 - \rho^2},$$

so that it depends on the design only through ρ . Check it against [Example \(A joint](#)

region for two coefficients), where the areas are 0.1197 and 0.1255 at $\rho = -0.639$. For which $|\rho|$ do the two regions have equal area there, and what happens to the ratio as the two regressors become collinear?

Solution. By [Theorem 12.2.1\(b\)](#) the ellipse has area $\pi \cdot 2F_\alpha s^2 \det(\mathbf{W})^{1/2}$, and writing ω_j for the diagonal entries of $\mathbf{W}$, $\det \mathbf{W} = \omega_1 \omega_2 (1 - \rho^2)$. The box has sides $2ts\sqrt{\omega_j}$, so area $4t^2 s^2 \sqrt{\omega_1 \omega_2}$; divide. In the example $F_{0.05}(2, 46) = 3.200$ and $t = 2.013$ give 1.240 at $\rho = 0$, and $1.240\sqrt{1 - \rho^2} = 0.954$ at $\rho = -0.639$, which is $0.1197/0.1255$. The areas agree when $\sqrt{1 - \rho^2} = 1/1.240$, that is $|\rho| = 0.592$. With uncorrelated estimates the ellipse is the larger region; as the regressors become collinear, $|\rho| \rightarrow 1$, the ratio tends to 0 and the ellipse becomes a thin needle across a large box, which the joint region mostly excludes.

Exercise B2.

In [Example \(A joint region for two coefficients\)](#), the hypothesis that one point of poverty and one point of single parenthood have the same coefficient is the line $\phi_1 = \phi_2$. Using [Lemma 12.2.3](#), show that this line meets the ellipse iff $|\hat{\phi}_1 - \hat{\phi}_2| \leq \sqrt{2F_\alpha(2, n - r)} \text{se}(\hat{\phi}_1 - \hat{\phi}_2)$. Compare this with the t test of $\phi_1 = \phi_2$. Which is appropriate if the hypothesis was formulated before seeing the data?

Solution. The line is $\{\phi_0 : \mathbf{a}^\top \phi_0 = 0\}$ with $\mathbf{a} = (1, -1)^\top$. By [Corollary 12.2.4](#), the values of $\mathbf{a}^\top \phi_0$ over the ellipse fill the interval $\mathbf{a}^\top \hat{\phi} \pm \sqrt{2F_\alpha} s \sqrt{\mathbf{a}^\top \mathbf{W} \mathbf{a}}$, so the line meets the ellipse iff this interval contains 0. The t test uses the smaller multiplier $t_{n-r, \alpha/2}$. It is the right test for a hypothesis fixed in advance. The ellipse criterion is the right one for a contrast chosen after looking at the estimates, because it holds simultaneously for all $\mathbf{a}$.

Exercise B3.

Let the columns of $\mathbf{\Lambda}$ lie in $\mathcal{C}(\mathbf{X}^\top)$ but be linearly dependent, with $\text{rank}(\mathbf{\Lambda}) = k < q$. Show that $\mathbf{W} = \mathbf{\Lambda}^\top \mathbf{G} \mathbf{\Lambda}$ is singular, that $\hat{\phi} - \phi \in \mathcal{C}(\mathbf{W})$ with probability one, and that

$$\left\{ \phi_0 \in \hat{\phi} + \mathcal{C}(\mathbf{W}) : (\hat{\phi} - \phi_0)^\top \mathbf{W}^{-} (\hat{\phi} - \phi_0) \leq k s^2 F_\alpha(k, n - r) \right\}$$

is an exact $1 - \alpha$ confidence set, a k -dimensional ellipsoid inside a k -dimensional affine subspace of $\mathbb{R}^q$.

Solution. $\mathbf{W} = (\mathbf{M}\mathbf{T})^\top (\mathbf{M}\mathbf{T})$ has rank $\text{rank}(\mathbf{M}\mathbf{T}) = \text{rank}(\mathbf{\Lambda}) = k$ (proof of [Corollary 7.5.3](#)). The vector $\hat{\phi} - \phi$ is $N_q(\mathbf{0}, \sigma^2 \mathbf{W})$, so it lies in $\mathcal{C}(\mathbf{W})$ almost surely ([Theorem 2.2.3](#)). By [Corollary 7.5.3](#) the quadratic form over $k s^2$ is $F(k, n - r)$ and does not depend on the generalized inverse. So the displayed set covers ϕ with probability $1 - \alpha$. Restricted to the affine subspace, write $\mathbf{W} = \mathbf{U}\mathbf{D}\mathbf{U}^\top$ with $\mathbf{U}$ ($q \times k$) orthonormal and $\mathbf{D}$ positive diagonal. For $\phi_0 - \hat{\phi} = \mathbf{U}\mathbf{c}$, the form is $\mathbf{c}^\top \mathbf{D}^{-1} \mathbf{c}$, an ellipsoid in the coordinates $\mathbf{c}$.

12.2.5.3 C. Going deeper

Exercise C1.

Let $\mathbf{X}$ have full column rank. Show that the confidence ellipsoid for $\boldsymbol{\beta}$ is a region of the form $\{\mathbf{b} : \ell_p(\mathbf{b}) \geq \ell_p(\hat{\boldsymbol{\beta}}) - k\}$, where $\ell_p(\mathbf{b}) = \max_{\sigma^2} \ell(\mathbf{b}, \sigma^2)$ is the profile log-likelihood of Chapter 7, and find k in terms of $F_\alpha(p, n-p)$. How does k compare with the value $\frac{1}{2}\chi_\alpha^2(p)$ suggested by the asymptotic theory of likelihood ratios?

Solution. The profile log-likelihood is $\ell_p(\mathbf{b}) = \text{const} - \frac{n}{2} \log \|\mathbf{Y} - \mathbf{X}\mathbf{b}\|^2$, and $\|\mathbf{Y} - \mathbf{X}\mathbf{b}\|^2 = \text{SSE} + (\hat{\boldsymbol{\beta}} - \mathbf{b})^\top \mathbf{X}^\top \mathbf{X} (\hat{\boldsymbol{\beta}} - \mathbf{b})$ by Eq. 6.3.1. So $\ell_p(\hat{\boldsymbol{\beta}}) - \ell_p(\mathbf{b}) = \frac{n}{2} \log\{1 + pF_{\mathbf{b}}/(n-p)\}$, where $F_{\mathbf{b}}$ is the statistic of Theorem 12.2.1(c). This is increasing in $F_{\mathbf{b}}$, so the ellipsoid is the region with $k = \frac{n}{2} \log\{1 + pF_\alpha(p, n-p)/(n-p)\}$. As $n \rightarrow \infty$, $pF_\alpha(p, n-p) \rightarrow \chi_\alpha^2(p)$ and $k \rightarrow \frac{1}{2}\chi_\alpha^2(p)$. For finite n , k is larger, and the exact region is larger than the asymptotic likelihood region.